ST. FROEBEL SENIOR SECONDARY SCHOOL

A-3 Block, Paschim Vihar, New Delhi-110063
CLASS - X (2026-27)
SUBJECT - Science

12A

Samarth

Time - 3 hour | Total Marks-80

GENERAL INSTRUCTIONS

Read the following instructions carefully and follow them:
(i)All questions are compulsory.
(ii) The question paper is divided into THREE sections -Biology, Chemistry and Physics.
Section A: Biology (30 Marks)
Section B: Chemistry (25 Marks)
Section C: Physics (25 Marks)

(iii) The question paper has MCQs, VSAS, SAS, LAs and C/S-BQs. Marks are given against each question.

(iv) There are case-based questions (CBQs) with three sub-questions and are of 4 marks each.

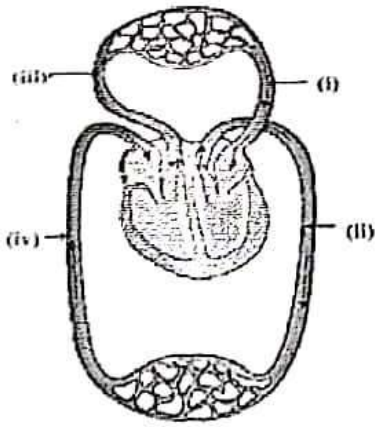
(v) Divide your answer sheet into three sections as per the question paper - Section A (Biology), Section (Chemistry) and Section C (Physics). It is compulsory to answer each question in its respective section. Do n mix answers from one section into the other section.

(vi) Instructions are given with each section and question, wherever necessary.

(vii) There is no overall choice in the question paper. However, an internal choice has been provided in a fe questions. Only one of the choices in such questions must be attempted.

Section A: BIOLOGY
MCQ 1 MARK EACH

Q1.Refer to the given schematic diagram of human double circulation



Labels in the Schematic Diagram:

(i) Pulmonary Vein (bringing oxygenated blood from lungs to left atrium)

(ii) Aorta / Systemic Artery (carrying oxygenated blood from left ventricle to body tissues)

(iii) Pulmonary Artery (carrying deoxygenated blood from right ventricle to lungs)

(iv) Vena Cava / Systemic Vein (bringing deoxygenated blood from body tissues to right atrium)

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A cardiac patient shows significantly reduced oxygen saturation in the left ventricle despite norm function. Blockage or abnormal pressure in which labeled vessel is most likely responsible for restr oxygenated blood inflow to the heart?

(a) Vessel (iii)

(b) Vessel (i)

(c) Vessel (iv)

(d) Vessel (ii)

Q2.A student added a specific neurotoxin to an isolated nerve-muscle preparation in a laboratory. Tl neurotoxin selectively blocks the functional action of **Receptor Z** without affecting the axon terminal, vesic fusion, or the breakdown of **Chemical Y.**

Axon Terminal

| [chemical X]

▼

(Vesicle) ⟶ Releases [Chemical Y] into Synaptic Cleft

|

▼

Binds to [Receptor Z] ⟶ Muscle Contraction

Which of the following outcomes will be observed during an electrical stimulation of the nerve?

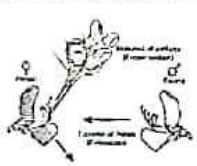
(a) Chemical Y will not be released into the synaptic cleft.

(b) Chemical Y will accumulate in the cleft, but muscle contraction will fail to occur.

(c) Muscle contraction will occur continuously without relaxation.

(d) An action potential will travel backward up the axon.

Q3. In an experimental study on flower structure, all stamens were removed from a bisexual flower befo pollen maturation. The flower was then covered with a protective bag. What will happen if compatible polle from another flower is manually dusted onto its stigma?





(a) No fertilization occurs because self-pollination was blocked.
(b) Fertilization proceeds normally, leading to seed formation.
(c) The ovary immediately degenerates without forming fruit.
(d) Only endosperm develops, but no embryo forms.

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Q4. Rohan eats a heavy meal rich in carbohydrates. Within an hour, his blood glucose concentration spikes temporarily before returning to normal baseline levels. Which physiological feedback sequence best explains how his body regulates this change?
A. The liver detects high glucose and secretes glucagon to break down glycogen into glucose.
B. The pituitary gland releases growth hormone to utilize the excess glucose for immediate muscular repair.
C. The pancreas detects elevated blood glucose and releases insulin, prompting cells and the liver to absorb and store glucose as glycogen.
D. The adrenal glands secrete adrenaline to burn off the extra carbohydrates through increased metabolic rate.

Q5. During a severe afternoon drought, a farmer notices that plants in the field begin to wilt, but simultaneousl the stomata in their leaves close rapidly to minimize excessive water loss through transpiration. Which plant hormone acts as the primary signal for this physiological response?
A. Auxin
B. Cytokinin
C. Gibberellin
D. Abscisic acid

Q6. A health educator is discussing different contraceptive strategies with a community group. She contrasts Method P (a mechanical device used externally that blocks gamete transfer) with Method Q (a mechanical device inserted surgically or clinically into the uterus, such as a Copper-T).
Based on your understanding of reproductive health and biological mechanisms,which of the following comparative analyses between Method P and Method Q is scientifically accurate?
A. Method Q acts as an external barrier that protects against sexually transmitted diseases (STDs) in addition t preventing unwanted pregnancies.
B. Method P releases copper ions to suppress sperm motility in the uterine cavity, whereas Method Q physicall intercepts semen at the time of coitus.
C. Method P prevents the meeting of sperm and ovum while also providing dual protection against STDs, whereas Method Q prevents implantation or fertilization via local uterine reactions but offers no protection against STDs.
D. Both Method P and Method Q function primarily by altering hormonal balances in the female body to inhib the release of eggs from the ovary.

Assertion & Reason Questions (1 Mark Each)

Directions: In questions 4 to 6, a statement of Assertion (A) is followed by a statement of Reason (R). Choo the correct option:
(a) Both (A) and (R) are true, and (R) is the correct explanation of (A).
(b) Both (A) and (R) are true, but (R) is NOT the correct explanation of (A).
(c) (A) is true, but (R) is false.
(d) (A) is false, but (R) is true.

Q7.Assertion (A): The inner lining of the small intestine possesses millions of microscopic finger-lil projections called villi.
Reason (R): Villi drastically increases the surface area available for the absorption of digested food into tl bloodstream.

Q8.Assertion (A): Electrical impulses are the fastest mode of signal transmission in the human body, but the cannot reach every single cell.
Reason (R): Nerve cells do not form direct physical connections with every individual cell, ~~necessitati~~ they use chemical messengers (hormones) for systemic regulation.

Q9.Assertion (A): Asexual reproduction produces offspring that are genetically identical clones of the pare organism.

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Reason (R): Asexual reproduction involves DNA replication without any major variations or fusion of ga

Short Answer Questions - I (2 Marks Each)

Q10.A person unintentionally consumes a substance that completely inhibits the enzyme pancreatic lipase ar pancreatic trypsin in the small intestine. Explain two (one each) direct physiological consequences th inhibition will have on the digestion and absorption of nutrients.

Q11.During a sudden stress response (e.g., encountering a dangerous animal),the body undergoes rap physiological changes.
Identify the gland and the specific hormone secreted into the bloodstream.
State two immediate bodily changes triggered by this hormone to prepare the organism for action.

Q12.Vegetative propagation is widely used in agriculture for growing crops like sugarcane and rose. State tw distinct advantages of this method over seed propagation for commercial growers.

Short Answer Questions - II (3 Marks Each)

Q13. Compare the structural and functional adaptations of the human lungs (alveoli) and the human kidney (nephrons) in executing their respective roles of exchange/filtration. Provide three distinct comparative points.

OR

Draw a well labeled diagram of the Human Respiratory System.

Q14. Name and briefly explain the mode of reproduction taking place in the following organisms:
1) Bryophyllum
2) Planaria
3) Rhizopus

Case-Based Question (4 Marks)

Q15. A microbiology lab monitored the reproduction rates of three unicellular/simple multicellular organism (Organism A, Organism B, and Organism C under controlled culture conditions:
Organism A: Divides symmetrically into two equal daughter cells in any plane (e.g., Amoeba).
Organism B: Breaks into distinct multicellular fragments when mature, each fragment developing into complete new individual (e.g., Spirogyra).
Organism C: Develops tiny outgrowths due to repeated cell division at one specific site; these outgrowtl detach upon maturing to live independently (e.g., Hydra).

1.Identify the exact mode of asexual reproduction exhibited Organism A, Organism B, and Organism C. (1.5 Marks)

2.How does the division in Leishmania differ structurally from the division process observed in Organism A? (1 Mark)

3.Explain why complex multicellular organisms (like humans or birds) cannot undergo reproduction through tl method observed in Organism B. (1.5 Marks)

Long Answer Question (5 Marks)

Q16.A) Describe the main events of the human menstrual cycle if fertilization does not occur followir ovulation. [1 Mark]

B) State the primary site of sperm production in the male reproductive system and name the hormor responsible for secondary sexual characteristics in males. [1mark]

C) Draw a neat, well-labeled diagram of the Female Reproductive System in humans, clearly labeling tl following parts: [3 marks]
Ovary

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iduct (Fallopian Tube)
Uterus
Cervix
Vagina
Endometrium

OR (CHOICE ONLY IN PART C)

Draw a neat, well-labeled diagram of the Male Reproductive System in humans, clearly labeling the followir parts: [3 marks]
Vas deference
Seminal vesicle
Prostate gland
Testis
Penis
Ureter

Section B: Chemistry

Question no 17-24 carries 1 mark each

Q 17. Marble statues are corroded or stained when they repeatedly come in contact with polluted rain water. Identify the main reason.



a) Decomposition of calcium carbonate to calcium oxide.
b) Polluted water is basic in nature hence it reacts with calcium carbonate.
c) Polluted water is acidic in nature hence it reacts with calcium carbonate.
d) Calcium carbonate dissolves in water to give calcium hydroxide.

Q 18. Neetu has two test tubes containing dilute hydrochloric acid and dilute sodium hydroxide solution, but they are not labelled. Adding which of the following solutions to the test tubes will help her visually identify th acidic and basic solution?
(a) only vinegar
(b) only baking soda
(c) only sodium chloride
(d) either vinegar or sodium chloride

Q 19. Sonia has aqueous solutions of three salts, sodium acetate, sodium chloride and ammonium chloride in three test tubes. The test tubes are not labelled. On checking, she finds the pH of the solutions to be 4.6, 7.0 an 8.9. Which of the following correctly matches the salts with their respective pH?

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	pH 4.6	pH 7.0	pH 8.9
A	sodium acetate	sodium chloride	ammonium chloride
B	sodium chloride	ammonium chloride	sodium acetate
C	ammonium chloride	sodium acetate	sodium chloride
D	ammonium chloride	sodium chloride	sodium acetate

a) A
b) B
c) C
d) D

Q 20. Select the oxidising agent for the following reaction:
$H_2S + I_2 \rightarrow 2HI + S$
(a) I_2
(b) H_2S
(C) HI
(d) S

Q 21. Sodium carbonate is a basic salt because it is a salt of a
(a) strong acid and strong base
(b) weak acid and weak base
(c) strong acid and weak base
(d) weak acid and strong base

Q 22. Nettle sting is a natural source of which acid?
(a) Methanoic acid
(b) Lactic acid
(c) Citric acid
(d) Tartaric acid

Q 23. Which of the following non-metal is lustrous?
(a) Sulphur
(b) Oxygen
(c) Nitrogen
(d) Iodine

Q 24 **Assertion (A):** Hydrogen gas is not evolved when a metal reacts with nitric acid.
Reason (R): Nitric acid is a strong oxidising agent.

a) Both A and R are true, and R is the correct explanation of A.
b) Both A and R are true, and R is not the correct explanation of A.
c) A is true but R is false.
d) A is false but R is true

Question no-25 carries 2 marks each

Q 25. Diana prepared a cake by two methods.

Method i) She added **baking soda** to the cake mixture and let the mixture stand for one hour before placing it i the oven to bake.

Method ii) She added **baking powder** to the cake mixture and let the mixture stand for one hour before placin it in the oven to bake.

State the difference in the cake mixtures that Diana is likely to have observed before baking. Explain why?

Question no 26 and 27 carries 3 marks each

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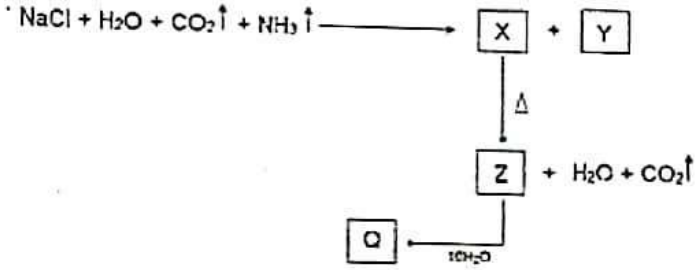
Q 26. Give reasons for the following:

(i) On strong heating, blue coloured copper sulphate crystals turn white.

(ii) Tarnished copper vessels are cleaned with tamarind juice.

(iii) Aluminium is more reactive than iron yet there is less corrosion of aluminium as compared to iron when both are exposed to air.

Q 27. (i) In the given series of reaction, name the compounds X, Y, Z, and Q



(ii) Which type of reaction is X to Z?

Question no 28 carries 5 marks

Q 28 During electrolysis of brine, a gas 'G' is liberated at anode. When this gas 'G' is passed through slaked lime, a compound 'C' is formed, which is used for disinfecting drinking water.

(i) Write the formulas of 'G' and 'C'. 1 mark

(ii) Which gas is liberated at cathode? 1 mark

(iii) State the chemical equations involved. 2 marks

(a) Electrolysis of brine

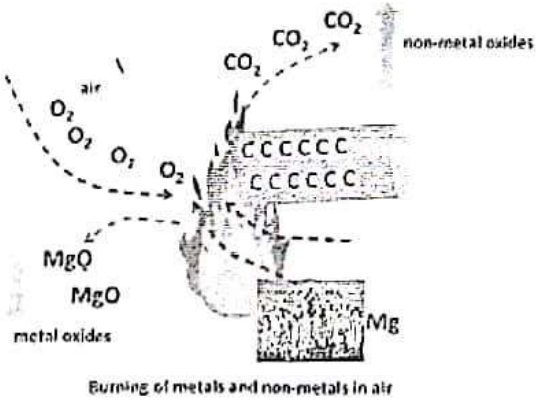
(b) Passing gas G through slaked lime.

(iv) What is the common name of compound 'C' ? Give its chemical name. 1 mark

Question no 29 carries 4 marks

Q 29 Almost all metals combine with oxygen to form metal oxides. Metal oxides are generally basic in nature. But some metal oxides show both basic as well as acidic behaviour. Different metals show different reactivities towards oxygen. Some react vigorously while some do not react at all.

Attempt A or B



Burning of metals and non-metals in air

A (a)What happens when copper is heated in air? (Give the equation of the reaction involved) 1 mark

(b) Why are some metal oxides categorized as amphoteric? 1 mark

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(c) Complete the following equations: 2 marks

(i) $Na_2O(s) + H_2O(l) \rightarrow$

(ii) $Al_2O_3 + 2NaOH \rightarrow$

OR

(B) On burning Sulphur in oxygen a colourless gas is produced.

(i) Write a chemical equation for the reaction. 1 mark

(ii) Name the gas formed. 1 mark

(iii) State the nature of the gas. 1 mark

(iv) What will be the action of this on a moist litmus paper? 1 mark

Section C: Physics

Q30.

Column I	Column I
P. Ray of light bends towards normal	1. $r < i$
Q. Substance having highest optical density	2. The ratio of refractive indices of the mediums
R. Slope of sin i vs sin r plot	3. Diamond

(a) P-1, Q-3, R-2
(b) P-1, Q-2, R-3
(c) P-2, Q-1, R-3
(d) P-3, Q-1, R-2

Q31. When you look at an object very close to your eyes, the : 1

(a) Ciliary muscles of your eye contract and the eye lens becomes thick.
(b) Ciliary muscles of your eye get relaxed and the eye lens becomes thick.
(c) Ciliary muscles of your eye contract and the eye lens becomes thin.
(d) Ciliary muscles of your eye get relaxed and the eye lens becomes thin.

Q32. Assertion (A): A point object is placed at a distance of 26 cm from a convex mirror of focal lengt 26 cm, the image will not form at infinity
Reason (R): For the above given system the equation $1/f = 1/v + 1/u$ gives v = infinity. 1

a) Both A and R are true and R is the correct explanation of A.
b) BothA and R are true and R is not the correct explanation of A.
c) A is true but R is false.
d) A is false but R is true.

Q33. A convex lens of focal length 40 cm is placed in contact with a concave lens at a focal length 25 cm What is the power of the combination ? 2

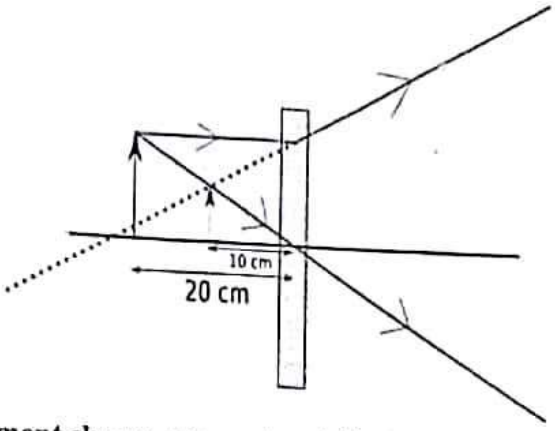
Q34. Draw a ray diagram to show the refraction of a Ray of light passing through an equilateral glas prism.Mark the angle through which the emergent ray bends from the direction of the incident ray an also name it. 2

OR

With the help of a ray diagram, explain any phenomenon of formation of spectrum in nature. 2

Q35. The below image shows the formation of an image with an optical instrument. 3

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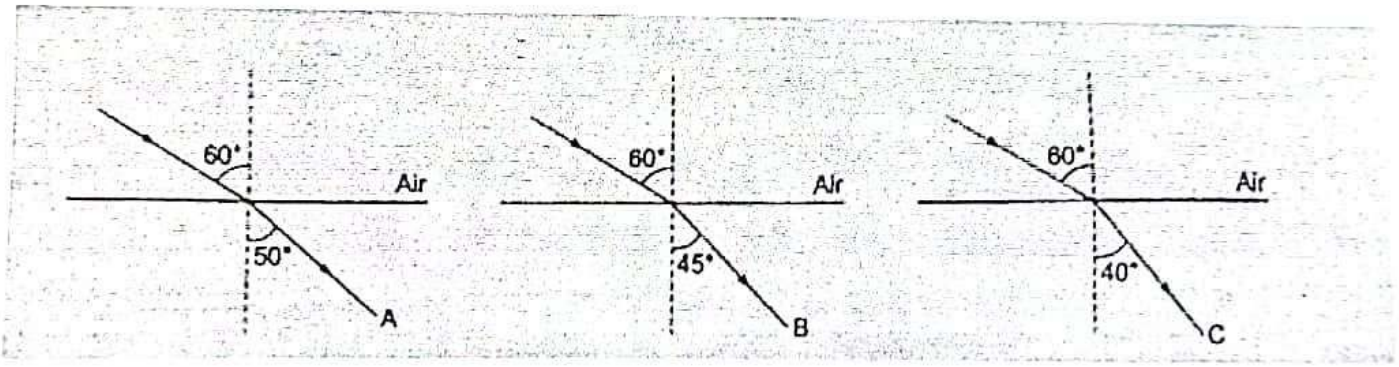
a. Identify the optical instrument shown as a rectangle in the image.
b. What type of images formed in this case?
c. Based on the measurements given in the image, calculate the focal length of the instrument.

Q36. Give reason for the following. 3

a. The colour of the clear sky is blue.
b. Twinkling of stars.
c. The pool appears to be less deep when viewed from outside.

Q37.Answer the following questions
(i) The refractive index of diamond is 2.42. What is the meaning of this statement? 1

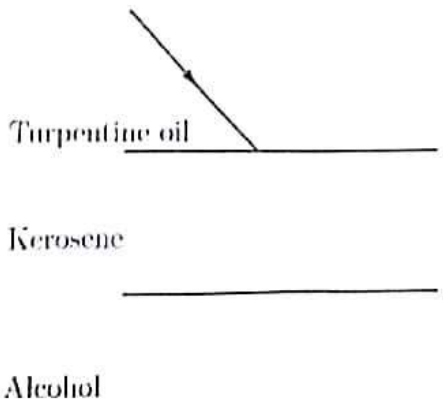
(ii) The diagram below represents the path of a light ray from three different media A, Band C.



In which of the following media is the speed of light maximum and why? 2

Or

(ii) Refractive index of turpentine oil, kerosene and alcohol are 1.47, 1.44 and 1.36 respectively. On th basis of this information, complete the following ray diagram to show the path of the ray of light throug each medium. 2



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Q38.During a school eye-screening programme, three students undergo a vision test. Their observatio are given below:

- Student A can read a book comfortably but cannot see the writing on the blackboard clearly from the la bench.
- Student B can see distant objects clearly but has difficulty reading a book when it is held at a norm reading distance.
- Student C, aged 45 years, finds it difficult to focus clearly on both nearby and distant objects. The doct explains that the power of accommodation of the eye has reduced considerably.

Based on the above information, answer the following:

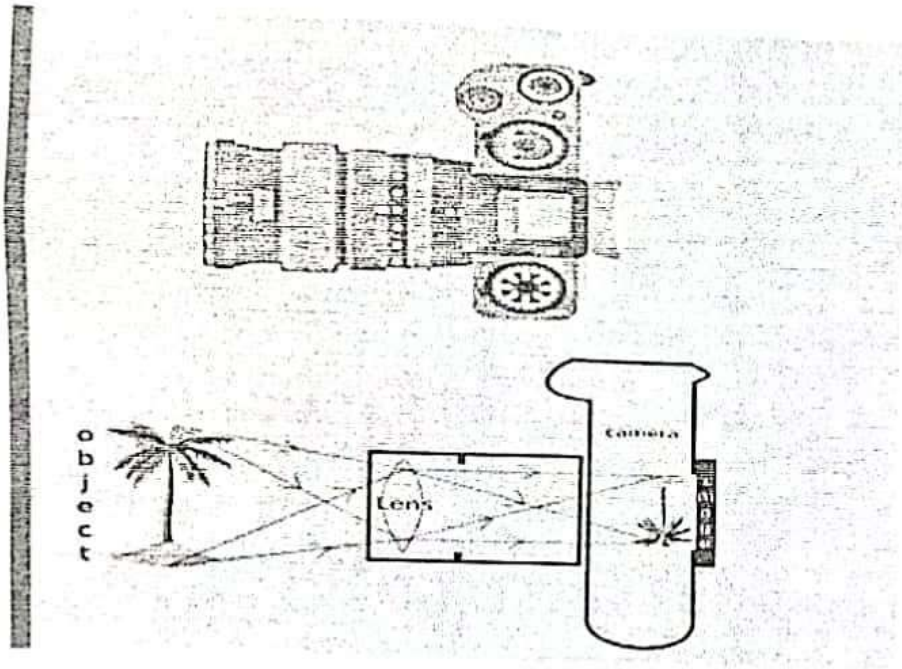
(a) Identify the defect of vision in Student A and draw the ray diagram depicting the correction of the defec Also state the causes of this defect. 4

Or

(a) Identify the defect of vision in Student B and draw the ray diagram depicting the correction of the defec Also state the causes of this defect.

(b) What is the likely defect in Student C? Explain why a combination of lenses may sometimes be prescribe to such a person.

Q39. The below image is that of a Digital Single Lens Reflector (DSLR) Camera which is used to tak high resolution photographs by professional photographers. The second image of the above two is schematic diagram of how an image is formed on the sensor of the camera. Based on your understandir of the lenses, answer the following questions. 5



(a) What type of lens is used in the DSLR camera shown in the image?

(b) What type of image is formed on the sensor?

(c) A photographer is using a DSLR camera with a lens of focal length f = 50 mm to take a close-up photograp of a small object. The lens projects an image onto the camera sensor that is located 60 mm behind the len Calculate the object distance (i.e., the distance between the object and the lens).

OR

(c) A photographer is using a DSLR camera to take a picture of a flower. The flower is positioned 150 m away from the camera lens. The actual height of the flower is 80 mm, and the image height formed on th camera's sensor is measured to be 20 mm. Calculate the focal length of the camera lens.

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